ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 4TH YEAR

MECHATRONIC SYSTEM DESIGN

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 4th year Mechatronic System Design paper 1 · practice a.

Written by Academic Central for practice, following final-year mechatronic design content at South African universities (e.g. University of Pretoria MOX 410 Design: ECSA GA3 engineering design, concept selection, analysis, manufacturing and costing; and MBB 410 Mechatronics: sensors, actuators and discrete-time control). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept other reasonable specifications, design choices and examples, and answers that differ only through rounding.

QUESTION 1

1.1 Problem definition and needs ✓; requirements and specifications ✓; concept generation and selection ✓; detailed design (analysis, calculations, drawings) ✓; build, test, verify and validate (and improve) ✓ (5)

1.2 A requirement says what the customer needs, often in words; a specification is a measurable value that the design must meet and that can be tested ✓✓
Any THREE measurable specifications, e.g. carries crates of up to 30 kg; top speed at least 1.5 m/s; runs at least 8 hours per charge; positions the crate within ±10 mm; stops within 0.5 m of an obstacle ✓✓✓ (5)

1.3 A: 0.3 × 7 + 0.4 × 8 + 0.3 × 6 = 7.1 ✓✓
B: 0.3 × 9 + 0.4 × 6 + 0.3 × 8 = 7.5 ✓
C: 0.3 × 6 + 0.4 × 9 + 0.3 × 6 = 7.2 ✓
Concept B has the highest weighted score, so it is chosen ✓✓ (6)

1.4 Verification: checking that the product meets its specifications ("did we build it right?") ✓, e.g. measuring that the robot reaches 1.5 m/s on a test track ✓
Validation: checking that the product meets the customer's real need ("did we build the right thing?") ✓, e.g. a trial in the real warehouse with the staff ✓ (4)

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QUESTION 2

2.1 Electrical: $v = R\,i + K_e\,\omega$ ✓✓
Mechanical: $J\,d\omega/dt = K_t\,i$ ✓✓ (4)

2.2 $i = (V - K_e\,\Omega)/R$, so $J\,s\,\Omega = K_t\,(V - K_e\,\Omega)/R$ ✓
$\Omega/V = K_t / (R\,J\,s + K_t\,K_e)$ ✓ $= 0.05 / (2 \times 10^{-4}\,s + 0.0025)$ ✓
$= 20 / (0.08\,s + 1)$ ✓
$K = 20$ rad/s per volt ✓ $\tau = 0.08$ s ✓ (6)

2.3 $\omega = V/K_e = 12 / 0.05 = 240$ rad/s ✓ $= 2\,291.83$ r/min ✓ (2)

2.4 $T_{stall} = K_t\,V / R = 0.05 \times 12 / 2$ ✓ $= 0.30$ N·m ✓ (2)

2.5 $P_{max} = ¼ \times T_{stall} \times \omega_{no\text{-}load}$ ✓ $= 0.25 \times 0.3 \times 240$ ✓ $= 18$ W ✓ (3)

2.6 About $4\tau = 0.32$ s ✓ (1)

2.7 The electrical equation gains an $L\,di/dt$ term, so the model becomes second order with an electrical time constant L/R ✓; it is usually much faster than the mechanical time constant and is often ignored ✓ (2)

[20]

QUESTION 3

3.1 $F = m(a + g \sin\theta + C_{rr} g \cos\theta)$ ✓ (4)
$= 40(0.5 + 9.81 \times 0.0872 + 0.02 \times 9.81 \times 0.9962)$ ✓✓
$= 40 \times 1.55 = 62.02$ N ✓

3.2 $T_{wheel} = F r / 2 = 62.02 \times 0.1 / 2$ ✓ $= 3.10$ N·m ✓ (2)

3.3 $\omega = v/r = 1.2 / 0.1 = 12$ rad/s ✓ $= 114.59$ r/min ✓ (2)

3.4 $P = T \omega = 3.10 \times 12$ ✓ $= 37.21$ W ✓ (2)

3.5 $N = 3\,000 / 114.59 = 26.18$ ✓, so choose about 26:1 (a standard ratio close to this, e.g. 25:1 or 27:1, is accepted) ✓ (2)

3.6 $T_{motor} = T_{wheel} / (N \eta)$ ✓ $= 3.10 / (26.18 \times 0.85)$ ✓ $= 0.14$ N·m ✓ (3)

3.7 Energy $= (2 \times 20 + 10) \times 4 = 200$ Wh ✓ (3)
$200 / 24 = 8.33$ Ah ✓
$8.33 / 0.8 = 10.42$ Ah (choose a 12 Ah battery) ✓

3.8 The real loads are uncertain (heavier crates, rough floors, worn parts, low battery voltage), and motors run hot near their limits ✓; a safety factor gives margin so the robot still works and the motors last ✓ (2)

[20]

QUESTION 4

4.1 $u[k] = u[k-1] + K_p (e[k] - e[k-1]) + K_i T e[k]$ ✓✓ (4)
It calculates only the change in output each sample, so wind-up is easy to limit and switching between manual and automatic control is bumpless ✓; it needs only the last values, not a growing sum ✓

4.2 $a = e^{(-0.01/0.08)} = e^{(-0.125)}$ ✓ $= 0.8825$ ✓ (5)
$K(1 - a) = 20 \times 0.1175 = 2.35$ ✓
$G(z) = 2.35 / (z - 0.8825)$ ✓✓

4.3 $y[k] = 0.8825\, y[k-1] + 2.35\, u[k-1]$ ✓✓ (6)
$y[1] = 0.8825 \times 0 + 2.35 \times 1 = 2.35$ ✓✓
$y[2] = 0.8825 \times 2.35 + 2.35 = 4.42$ ✓✓

4.4 All the poles must lie inside the unit circle in the z-plane (magnitude less than 1) ✓✓ (2)

4.5 Rise time $\approx 2.2 \times 0.08 = 0.18$ s ✓ (3)
About 10 to 20 samples per rise time are needed ✓
So T of about 0.01 s to 0.02 s is suitable; 0.01 s is a good choice ✓

4.6 When the actuator saturates (e.g. 100% PWM), the integral term keeps growing while the error stays large ✓; when the error reverses, the large integral causes a big overshoot ✓ (3)
Prevent it by stopping (clamping) the integration while the output is saturated, or by back-calculation ✓

4.7 At low speed only a few pulses (or none) arrive in each sample, so the speed jumps between a few values ✓
Measure the time between pulses instead, use a higher-resolution encoder, or average over a longer window ✓ (2)

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QUESTION 5

5.1 FMEA lists each way a part can fail, the effect and the cause, and rates its severity, occurrence and detection, so the biggest risks are fixed first ✓✓
RPN = S × O × D = 8 × 3 × 5 ✓ = 120 ✓ (4)

5.2.1 R = 0.95 × 0.98 × 0.99 ✓ = 0.92 ✓ (2)

5.2.2 R = 1 − (1 − 0.9)² ✓ = 0.99 ✓ (2)

5.2.3 R(t) = e^(−t/MTBF) = e^(−500/2 000) ✓ = 0.78 ✓ (2)

5.3 Any THREE: separate power and signal grounds, joined at one point (star ground); keep motor wires away from sensor wires and twist them; flyback diodes and snubbers on inductive loads; decoupling capacitors next to each IC; shielded cables for sensors; a separate or filtered supply for the logic ✓✓✓ (3)

5.4 DFMA designs parts so they are cheap and easy to make and assemble ✓, e.g. using fewer parts, snap-fits instead of screws, standard fasteners, or parts that can only fit one way ✓ (2)

[15]

TOTAL: 100

END OF MARKING GUIDELINE